

PRODUCT REQUIREMENT DOCUMENT (PRD)

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1. Project Overview	3
2. Background & Business Objective	3
3. Product Purpose	4
4. System Functionalities	4
5. User Stories & Use Cases	6
6. Features Included (Scope Definition)	7
7. Features Not Included (Scope Control)	8
8. Assumptions & Constraints	8
9. Risks & Questions Throughout Development	9

Project Name: FinSight AI — SME Financial Health Advisor

Version: 1.0

1. Project Overview

- **Problem Statement:** Many Malaysian SME owners handle their financial documents such as invoices, receipts, and bank statements manually. Without a dedicated finance team, it is difficult to maintain a clear and real-time view of cash flow, overdue invoices, and rising expenses.
- **Target Domain:** Financial health monitoring and decision support for Malaysian SMEs across F&B, automotive, personal-care, and other small-retail sectors.
- **Proposed Solution Summary:** FinSight AI converts uploaded financial documents into clear, ringgit-based decisions. Users can upload PDFs, invoices, or receipt images, and the system will extract transactions, forecast cash flow, detect risks, and generate ranked recommendations. All outputs are explained in plain English using a Malaysian large language model (Ilmu AI).

2. Background & Business Objective

- **Background of the Problem:** Financial information in SMEs is often scattered across physical documents, WhatsApp images, and spreadsheets. Owners without financial training typically spend limited time reviewing this information, and insights are usually derived from monthly bookkeeping rather than real-time monitoring.
- **Importance of Solving This Issue:** Cash flow issues are difficult to recover from once they are reflected in the bank balance. A system that identifies risks earlier using existing data allows business owners to take action sooner and maintain more control over their financial situation.

- **Strategic Fit / Impact:** FinSight AI uses Ilmu AI, a Malaysian-developed large language model, as its reasoning engine. This ensures that the natural language explanation layer is powered by a local AI solution rather than a foreign service.

3. Product Purpose

3.1. Main Goal of the System: The system aims to provide Malaysian SME owners with an AI-assisted financial advisor that transforms uploaded financial documents into actionable, ringgit-based insights without requiring accounting expertise.

3.2. Intended Users (Target Audience):

- Independent café and bakery owners
- Independent auto repair workshop owners
- Pet grooming and boarding operators
- Small retailers and service providers across F&B, automotive, and personal-care sectors
- Malaysian SMEs without a dedicated finance team

4. System Functionalities

4.1. Description: FinSight AI is a generative financial co-pilot designed for SME owners in high frequency transaction industries. The system operates as a high-speed intelligence engine that transforms fragmented financial data like fluctuating supplier costs and multi-platform revenue into an unified and actionable UI. Using a Prompt-to-Insight model, it captures complex relationships between cash flow, outstanding invoices and more to provide “human-in-the-loop” financial management.

4.2. Key Functionalities:

- **Document Ingestion and Parsing:** A dedicated “Upload Documents” module for high-frequency receipt and invoice processing and advanced OCR and parsing logic to extract line-item data from messy physical documents or images.
- **Cash Flow Forecast:** Utilises historical transaction trends to generate an predictive model and calculates future liquidity based on current balance and anticipated platform payouts.
- **Risk Detection and Alert:** Scanning detector for financial anomalies and risks and triggers alerts when projected balances dip below safety thresholds.
- **AI Decision Recommendations:** Powered by the Z AI engine to turn data into strategy and provides actionable insights like suggesting steps or strategies to overcome certain issues.
- **Financial Dashboard:** The central command area providing a high-level overview of the forecasting cash flow, risks and alerts, AI recommendations and more.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: The system adopts Z AI GLM as the primary large language model, integrated through an LLM-as-a-Service API. It is selected as it supports the system’s core features and its ability to process unstructured inputs makes it aligned for transforming raw user data into meaningful insights required by the system. Z AI GLM demonstrates robust natural language understanding and context-aware reasoning. This enables it to interpret free-text inputs and identify related patterns for accurate risk classification, forecasting generation and financial advice. Furthermore, the model’s high adherence to instruction-based prompts ensures that the system produces consistent outputs that align with the requirements of the Risk and Recommendation services. Its API-based deployment ensures smooth integration and scalability with the Prompt Builder and Context Manager components.

4.3.2. Prompting Strategy: The team utilises a Multi-Step Structured Prompting strategy. Rather than executing a single exhaustive request, the system breaks the analysis into sequential API calls. This approach fits the financial use case by narrowing the model's focus to specific categories like separating immediate debt collection from long-term financing in each step. By segmenting the instructions, the system reduces the likelihood of logic errors and ensures that the AI-driven recommendations are highly specialised and accurately referencing Malaysian-specific entities while maintaining high logical consistency across different recommendation categories.

4.3.3. Context & Input Handling: The system manages unstructured or oversized inputs through a specialised Context Manager and Token Manager pipeline. To prevent processing errors from data that exceeds the model's context window, the system employs the data pre-processing, intelligent prioritisation and structured truncation.

4.3.4. Fallback & Failure Behavior: The product ensures data integrity and a graceful error state through a multi-layered validation path like post-processing, schema validation and service resilience.

5. User Stories & Use Cases

- **User Stories:**

- As an SME owner, I want to upload my bank statements and supplier invoices so that I can view my cash balance without manually entering data into Excel.
- As an SME owner, I want the system to detect risks early so that I can act before issues affect my bank balance.
- As a non-finance user, I want each risk to be explained clearly in plain English with specific Ringgit amounts so that I can fully understand the situation.
- As a business owner, I want ranked recommendations so that I can focus only on the most important actions

- **Use Cases (Main Interactions):**

- **Upload-to-Insight Flow:** The user uploads a PDF bank statement or image document (JPG/PNG receipt) via the Upload Documents screen. The AI parser automatically extracts transaction line items and structures them into transaction records. Extracted data is immediately reflected in the Dashboard, updating the current balance and the Recent Transactions feed.
- **Cash Flow Forecast Flow:** The user opens the Cash Flow Forecast screen to view an 8-week projection built from historical transaction patterns. Weekly rows show Projected Inflow, Projected Outflow, Net, Projected Balance, and a SAFE / CASH GAP status. A summary banner flags the specific week and amount of the projected minimum balance. An inline recommendation prompt at the bottom of the page invites the user to see how following the top recommendations can close the projected gap.
- **Risk Detection Flow:** The user navigates to the Risk Alerts screen. The deterministic rule engine scans the transaction history and applies five rules (cash flow gap, overdue invoices, expense spike, revenue decline, large payable). Ilmu AI generates plain-English explanations for each detected risk, stating the specific Ringgit amount affected. The system displays severity-coded alerts (Critical / High / Medium / Low) along with an overall Financial Risk Score (0–10) and a next-review date.
- **Recommendation Flow:** After reviewing the identified risks, the user navigates to the Recommendations screen. Ilmu AI processes a financial snapshot that includes balance, overdue invoices, upcoming expenses, and key cost areas, and generates up to six ranked actions across four categories. Each recommendation includes a difficulty rating, estimated timeframe, projected RM impact, and three to five concrete action steps. Where relevant, a linked invoice reference is also provided. The user can mark an action as completed by selecting “Mark as Actioned.”

6. Features Included (Scope Definition)

- Document ingestion and parsing supports PDF bank statements, invoices, and JPG or PNG receipts up to 15 MB per file. The system extracts structured transaction data automatically.
- Cash flow forecasting provides an 8-week projection based on historical inflow and outflow patterns, including weekly visualisation and cash gap indicators.
- Risk detection and alerts are based on five deterministic rules, including cash flow gaps, overdue invoices, expense spikes, revenue decline, and large payables. Each risk is scored and explained clearly.
- AI decision recommendations include four to six prioritized actions across collections, supplier management, cost optimization, and financing. Each recommendation includes financial impact estimates and actionable steps.
- Financial dashboard presents a unified view of current balance, monthly revenue and expenses, outstanding invoices, and integrated insights from risk detection, forecasting, and recommendations.

7. Features Not Included (Scope Control)

- The system does not include direct bank integration. All data must be uploaded manually, and no Open Banking or PayNet APIs are used.
- The system does not perform tax calculations or SST filing. It focuses only on financial health monitoring.
- The system does not include a native mobile application. It is delivered as a Flutter web application.
- The system does not execute actions automatically. It only provides recommendations, and users must carry out the actions independently.

8. Assumptions & Constraints

- **LLM Cost Constraint:** Each user session is estimated to consume approximately 8000 tokens across all Z.AI API calls.
- **Technical Constraints:** System is web-based via Flutter; File uploads are restricted to PDF, JPG and PNG formats, capped at 15 MB per file.

- **Performance Constraints:** Complex financial snapshots may push close to the 8000 token limit, resulting in response times of up to 3 minutes; Inputs are summarised and capped at a fixed length before each AI call to prevent token overflow.
- **User Input:** SME owners must manually upload documents to trigger analysis (no auto-ingestion or background polling); Z.AI outputs require human review before acting as GLM responses may not be arithmetically precise.

9. Risks & Questions Throughout Development

Design Similarity

Risk: Since FinSight AI generates financial recommendations rather than visual designs, the main risk is that different SME users may receive similar or generic advice even when their financial situations are different. If this happens, users may feel that the system is giving standard financial tips instead of personalised, data-driven guidance.

Question: How can we ensure that AI-generated recommendations remain specific to each SME's uploaded documents, transaction history, detected risks, and cash flow situation?

Response: The system reduces this risk by generating recommendations from each business's own financial snapshot, including current balance, overdue invoices, upcoming expenses, cash flow forecast, and detected risks. The AI prompt also requires specific Ringgit amounts, invoice references, timeframes, and action steps so that outputs are tied to the user's actual data instead of broad financial advice.

Frame Stability

Risk: The core screens of FinSight AI, including Dashboard, Transactions, Upload Documents, Risk Alerts, Cash Flow Forecast, and Recommendations, may display dynamic AI-generated content that varies in length. Long explanations, vendor names, invoice references, or large Ringgit values could affect card spacing, readability, or layout consistency.

Question: Does the user need to manually fix the layout if the interface becomes visually broken or difficult to read after AI output is displayed?

Response: Users should not need to manually fix the layout. FinSight AI uses fixed application views rather than editable design frames. The frontend displays AI outputs through structured fields and reusable UI components, while backend responses follow a consistent JSON format. A future improvement is to add visual regression testing to ensure all screens remain stable when displaying unusually long or complex financial data.

Drawing Interpretation

Risk: FinSight AI depends on uploaded documents such as receipts, invoices, and bank statements. If OCR or document parsing fails, the AI may receive incomplete or unclear text, which could lead to incorrect classification, missing transaction data, or unreliable financial insights.

Question: What happens if the AI is unable to contextualise or understand the uploaded document image, receipt, invoice, or bank statement?

Response: FinSight AI does not rely on the AI model to directly interpret raw images or visual layouts. The system first uses OCR or document parsing to extract text before sending the cleaned text to the AI model. If the uploaded document cannot be understood, it should be marked as failed or unknown instead of allowing unreliable data to affect forecasts or recommendations. A future improvement is to provide a manual correction flow so users can review extracted amounts, dates, and transaction types before the data is used in risk detection.